

mb-pc40s-2018-jun

Typst conversion review. OCR content and mathematical accuracy require editorial review.

Question 1

Pierre pushes his car into a garage. The radius of a tire on his car is 22 cm . Determine the distance travelled by his car if the tire rotated a total of 1000° .

Solution

$$\begin{aligned}\theta &= (1000)\left(\frac{\pi}{180}\right) \\ &= \frac{50\pi}{9}\end{aligned}$$

1 mark for conversion

$$\begin{aligned}s &= \theta r \\ s &= \left(\frac{50\pi}{9}\right)(22) \\ &= \frac{1100\pi}{9} \text{ cm}\end{aligned}$$

1 mark for substitution

2 marks

$$s = 383.972 \text{ cm}$$

Question 2

Solve, algebraically.

$$7^{\frac{x}{2}} = 85$$

Solution

$$\frac{x}{2} \log 7 = \log 85$$

$$x \log 7 = 2 \log 85$$

$$x = \frac{2 \log 85}{\log 7}$$

$$x = 4.566142$$

$$x = 4.566$$

$\frac{1}{2}$ mark for applying logarithms

1 mark for power law

$\frac{1}{2}$ mark for evaluating quotient of logarithms

2 marks

Question 3

Solve, algebraically, over the interval $[0, 2\pi)$.

$\sin x(\sec x + 3) = 0$

Solution

$\sin x = 0$	$\sec x = -3$	$\frac{1}{2}$ mark for solving for $\sin x$ $\frac{1}{2}$ mark for solving for $\sec x$
	$\cos x = -\frac{1}{3}$	1 mark for reciprocal
	$x_r = 1.230959$	
$x = 0, \pi$	$x = 1.911, 4.373$	2 marks for values of x (1 mark for each branch)
		4 marks

Question 4

"

Brahim invests \$2500 at an annual interest rate of 6.75% compounded monthly. Determine, algebraically, how many years it will take for his investment to reach an amount of \$10500.

Use the formula:

$$A = P \left(1 + \frac{r}{n} \right)^{nt}$$

where A = the amount of the investment after t years

P = the principal of the investment

r = the annual interest rate (as a decimal)

n = the number of compounding periods per year

t = the length of the investment in years

Solution

$$10500 = 2500 \left(1 + \frac{0.0675}{12} \right)^{12t}$$

$$4.2 = (1.005625)^{12t}$$

$$\log 4.2 = 12t \log 1.005625$$

$$\frac{\log 4.2}{12 \log 1.005625} = t$$

$$21.320250 = t$$

$$21.320 \text{ years} = t$$

$\frac{1}{2}$ mark for substitution

$\frac{1}{2}$ mark for simplification

$\frac{1}{2}$ mark for applying logarithms

1 mark for power law

$\frac{1}{2}$ mark for evaluating quotient of logarithms

3 marks

Question 5

There are 13 adults and 18 children who can be selected to go on a trip. Determine the number of ways 4 adults and 7 children can be selected if Sandra, one of the adults, must be selected.

Solution

${}_1C_1 \cdot {}_{12}C_3 \cdot {}_{18}C_7$	1 mark for ${}_{12}C_3$ $\frac{1}{2}$ mark for ${}_{18}C_7$
7001280	$\frac{1}{2}$ mark for product of combinations
	2 marks

Note:

- ${}_1C_1$ does not need to be shown.

Question 6

(2417)

In the binomial expansion of $\left(\frac{2}{x^2} - x^3\right)^9$, determine and simplify the 6th term.

Note: A calculator is not required for the remaining test questions.

Solution

$$t_6 = {}_9C_5 \left(\frac{2}{x^2}\right)^4 (-x^3)^5 \quad 2 \text{ marks (1 mark for } {}_9C_5; \frac{1}{2} \text{ mark for each consistent factor)}$$

$$t_6 = 126 \left(\frac{16}{x^8}\right) (-x^{15})$$

$$t_6 = -2016x^7 \quad 1 \text{ mark for simplification (} \frac{1}{2} \text{ mark for coefficient; } \frac{1}{2} \text{ mark for exponent)}$$

3 marks

Question 7

Given that $f(x) = \{(-1, 0), (0, 2), (1, -3), (2, 4)\}$, evaluate $f(f(0))$.

Solution

$$\begin{array}{rcl} f(f(0)) & & \\ f(2) & 1/2 \text{ mark for } f(0) = 2 & \\ 4 & 1/2 \text{ mark for } f(f(0)) & \\ & 1 \text{ mark} & \end{array}$$

Question 8

The point $(-\frac{5}{6}, b)$ is on the unit circle and is in quadrant III.

Determine the exact value of b .

Solution

$$\left(-\frac{5}{6}\right)^2 + b^2 = 1 \quad 1/2 \text{ mark for substitution}$$

$$b^2 = 1 - \frac{25}{36}$$

$$b^2 = \frac{11}{36}$$

$$b = \pm \frac{\sqrt{11}}{6}$$

$$b = -\frac{\sqrt{11}}{6} \quad 1/2 \text{ mark for the exact value of } b$$

1 mark

Question 9

Given the following row of Pascal's Triangle, determine the values of the next row.

1 6 15 20 15 6 1

Solution

1 7 21 35 35 21 7 1

Question 10

The following transformations are applied to $f(x)$, resulting in a new function, $g(x)$.

- reflection over the x -axis
- vertical stretch by a factor of 3
- horizontal stretch by a factor of 4

State the equation of $g(x)$ in terms of $f(x)$.

$$g(x) =$$

Solution

$$g(x) = -3f\left(\frac{1}{4}x\right)$$

1 mark for vertical reflection

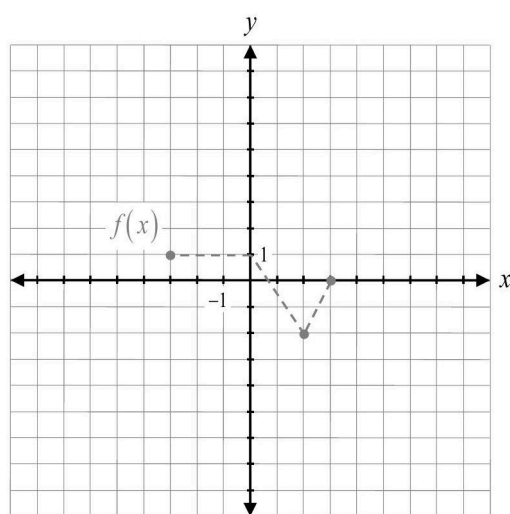
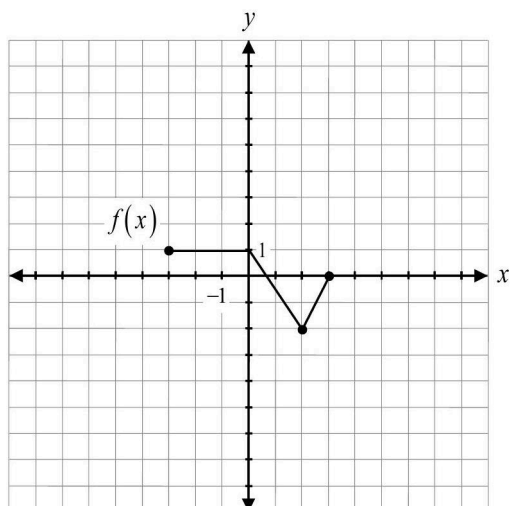
1 mark for vertical stretch

1 mark for horizontal stretch

3 marks

Question 11

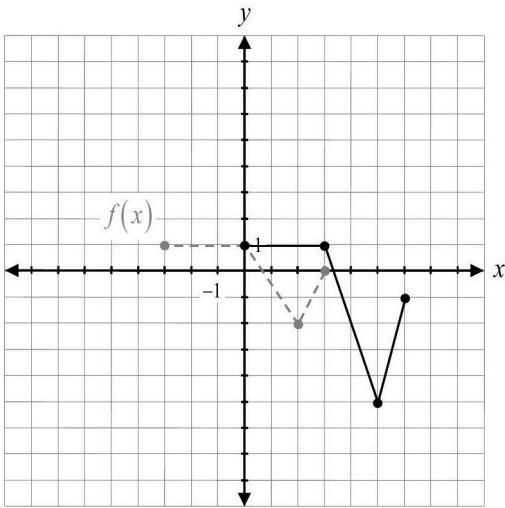
Given the graph of $f(x)$, sketch the graph of $y + 1 = 2f(x - 3)$.



The graph of
 $f(x)$ has
already been
drawn for your
reference.

No marks will
be awarded for
the graph of
 $f(x)$

Solution



- 1 mark for vertical stretch
- 1 mark for horizontal translation
- 1 mark for vertical translation
- 3 marks

Question 12

State the equation of the horizontal asymptote of $f(x) = \frac{2x^2-3x+5}{4x^2+2x-7}$.

Solution

$$y = \frac{1}{2}$$

1 mark

Question 13

Given that $(x + 4)$ is one of the factors of $p(x) = x^3 + 6x^2 - 32$, express $p(x)$ in completely factored form.

$p(x) =$

Solution

-4	1	6	0	-32
	↓	-4	-8	32
	1	2	-8	0

$\frac{1}{2}$ mark for $x = -4$

1 mark for synthetic division (or any equivalent strategy)

$$p(x) = (x + 4)(x^2 + 2x - 8)$$
$$p(x) = (x + 4)(x + 4)(x - 2)$$

$\frac{1}{2}$ mark for consistent product of factors

or

2 marks

$$p(x) = (x + 4)^2(x - 2)$$

Question 14

Prove the identity for all permissible values of θ .

$$\frac{2 \cos^2 \theta}{1 - \cot \theta} = \frac{\sin 2\theta}{\tan \theta - 1}$$

Left-Hand Side	Right-Hand Side

Solution

Method 1

Left-Hand Side	Right-Hand Side
$\frac{2 \cos^2 \theta}{1 - \frac{\cos \theta}{\sin \theta}}$ $\frac{2 \cos^2 \theta}{\frac{\sin \theta - \cos \theta}{\sin \theta}}$ $\frac{2 \cos^2 \theta \sin \theta}{\sin \theta - \cos \theta}$	$\frac{2 \sin \theta \cos \theta}{\frac{\sin \theta}{\cos \theta} - 1}$ $\frac{2 \sin \theta \cos \theta}{\frac{\sin \theta - \cos \theta}{\cos \theta}}$ $\frac{2 \sin \theta \cos^2 \theta}{\sin \theta - \cos \theta}$

1 mark for correct substitution of identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Method 2

Left-Hand Side	Right-Hand Side
$\frac{2 \cos^2 \theta}{1 - \cot \theta}$	$\frac{2 \sin \theta \cos \theta}{\frac{\sin \theta}{\cos \theta} - 1}$ $\frac{2 \sin \theta \cos \theta}{\frac{\sin \theta - \cos \theta}{\cos \theta}}$ $\frac{2 \sin \theta \cos^2 \theta}{\sin \theta - \cos \theta}$ $\frac{\sin \theta (2 \cos^2 \theta)}{\sin \theta (1 - \frac{\cos \theta}{\sin \theta})}$ $\frac{2 \cos^2 \theta}{1 - \cot^2 \theta}$

1 mark for correct substitution of identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Method 3

Left-Hand Side	Right-Hand Side
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$\frac{2 \cos^2 \theta}{1 - \cot \theta}$ $\frac{2 \cos^2 \theta}{1 - \frac{1}{\tan \theta}}$ $\frac{2 \cos^2 \theta}{\frac{\tan \theta - 1}{\tan \theta}}$ $\frac{2 \cos^2 \theta}{\tan \theta - 1} (\tan \theta)$ $\frac{2 \cos^2 \theta \left(\frac{\sin \theta}{\cos \theta} \right)}{\tan \theta - 1}$ $\frac{2 \cos \theta \sin \theta}{\tan \theta - 1}$ $\frac{\sin 2\theta}{\tan \theta - 1}$	$\frac{\sin 2\theta}{\tan \theta - 1}$
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1 mark for algebraic strategies

1 mark for logical process to prove the identity

1 mark for correct substitution of identities

3 marks

Question 15

Restaurant A has 5 types of hamburgers, 2 types of french fries, and 10 types of drinks.

Restaurant B has 4 types of hamburgers, 5 types of french fries, and 6 types of drinks.

If a meal is made up of a hamburger, french fries, and a drink, justify which restaurant offers a greater variety of meals.

Solution

Restaurant A: $5 \cdot 2 \cdot 10 = 100$ meals

Restaurant B: $4 \cdot 5 \cdot 6 = 120$ meals

Restaurant B offers a greater variety of meals.

Question 16

Express $\log_7(2x - 5) + 2\log_7 3$ as a single logarithm.

Solution

$$\begin{aligned} \log_7(2x - 5) + \log_7 3^2 & \quad 1 \text{ mark for power law} \\ \log_7(2x - 5) + \log_7 9 & \\ \log_7(9(2x - 5)) & \quad 1 \text{ mark for product law} \\ \text{or} & \quad 2 \text{ marks} \\ \log_7(18x - 45) & \end{aligned}$$

Question 17

Explain why $11!$ is not the total number of 11-letter arrangements that can be made from the word CELEBRATION.

Solution

The total number of possible arrangements is half of $11!$ because switching the two Es creates duplicate arrangements.

1 mark

Question 18

Given $f(x) = x - 1$, identify a point on the graph of $y = \sqrt{f(x)}$.

A.

$(0, -1)$

B.

$(3, 2)$

C.

$(1, 0)$

D.

$(0, 1)$

Solution

Correct answer: C

$(1, 0)$

Question 19

Identify the total number of possible arrangements of 6 adults and 4 children seated in a row if the children must sit together.

A.

$6!4!$

B.

$7!4!$

C.

$10!$

D.

$6!$

Solution

Correct answer: B

$7!4!$

Question 20

Identify the exact value of $\sec\left(-\frac{7\pi}{3}\right)$.

A.

$$-2$$

B.

$$-\frac{2}{\sqrt{3}}$$

C.

$$\frac{2}{\sqrt{3}}$$

D.

$$2$$

Solution

Correct answer: D

$$2$$

Question 21

Given $\log_x\left(\frac{1}{25}\right) = -2$, identify the value of x .

A.

-5

B.

$-\frac{1}{5}$

C.

$\frac{1}{5}$

D.

5

Solution

Correct answer: D

5

Question 22

Identify the equation for all of the asymptotes on the graph of $y = \tan x$.

A.

$$x = k\pi, k \in \mathbb{Z}$$

B.

$$x = 2k\pi, k \in \mathbb{Z}$$

C.

$$x = \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$$

D.

$$x = \frac{\pi}{2} + 2k\pi, k \in \mathbb{Z}$$

Solution

Correct answer: C

$$x = \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$$

Question 23

If $p(x) = 3(m)(x + 1)^2$ is a cubic function with a y -intercept of -12 , identify the missing factor, m .

A.

$$m = x - 4$$

B.

$$m = x + 4$$

C.

$$m = x + 12$$

D.

$$m = x - 12$$

Solution

Correct answer: A

$$m = x - 4$$

Question 24

Identify the number of negative terms in the binomial expansion of $(x - y)^5$.

A.

2

B.

3

C.

5

D.

6

Solution

Correct answer: B

3

Question 25

Given $f(x) = x^2$, identify which equation represents the graph of $y = f(x)$ after a translation of 5 units to the left.

A.

$$y = (x + 5)^2$$

B.

$$y = (x - 5)^2$$

C.

$$y = x^2 - 5$$

D.

$$y = x^2 + 5$$

Solution

Correct answer: A

$$y = (x + 5)^2$$

Question 26

When a polynomial, $p(x)$, is divided by $(x - 7)$, the remainder is 24 . Identify the only statement that must be true.

A.

$x = 7$ is a zero of $p(x)$

B.

$p(7) = 24$

C.

$x = 24$ is a zero of $p(x)$

D.

the y -intercept is 24

Solution

Correct answer: B

$p(7) = 24$

Question 27

Given $f(x) = \frac{(2x+1)(x-8)}{(x-8)(x+4)}$, state the equation(s) of the vertical asymptote(s).

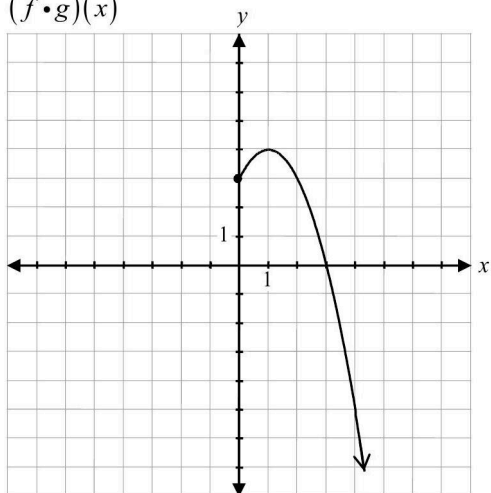
Solution

$$x = -4 \quad 1 \text{ mark}$$

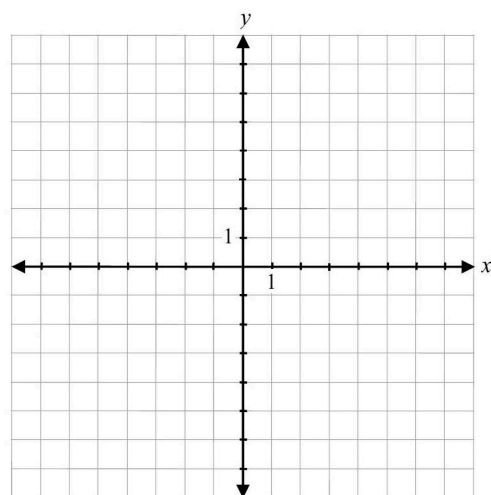
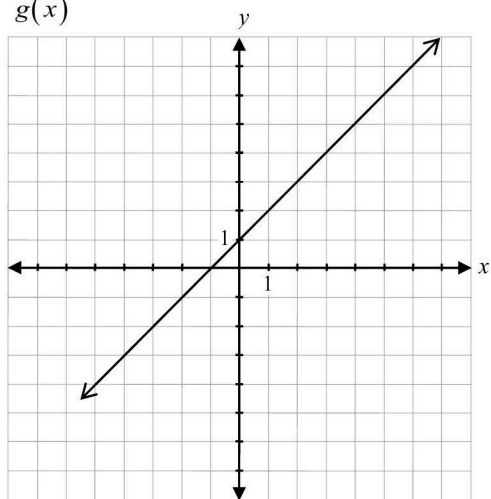
Question 28

Given the graph of $(f \cdot g)(x)$ and $g(x)$, sketch the graph of $f(x)$.

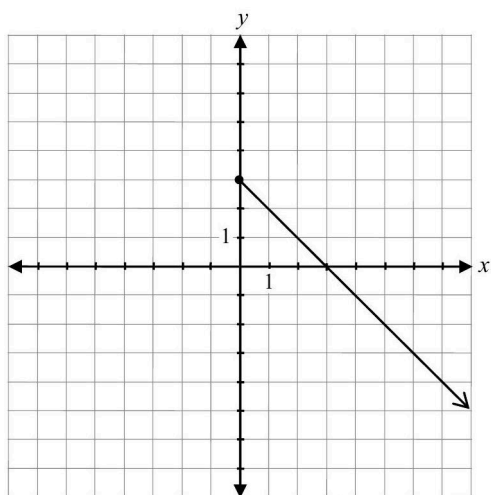
$(f \cdot g)(x)$



$g(x)$



Solution



1 mark for operation of division

1 mark for restricted domain

2 marks

Question 29

Brian was asked to state the zeros of the polynomial $p(x) = (x + 2)(x - 5)(x - 1)$.

Brian's response:

Zeros: $(x + 2)$, $(x - 5)$, $(x - 1)$

Explain why his response is incorrect.

Solution

Brian stated the factors instead of the zeros.

Question 30

Simplify ${}_{n+3}C_2$.

Solution

$$\frac{(n+3)!}{(n+3-2)!2!}$$

$$\frac{(n+3)!}{(n+1)!2!}$$

$$\frac{(n+3)(n+2)(n+1)!}{(n+1)!2}$$

$$\frac{(n+3)(n+2)}{2}$$

or

$$\frac{n^2+5n+6}{2}$$

$\frac{1}{2}$ mark for substitution

1 mark for factorial expansion

$\frac{1}{2}$ mark for simplification of factorials

2 marks

Question 31

Verify that the equation $2 \cos^2 x = \sin x + 1$ is true for $x = \frac{\pi}{6}$.

Solution

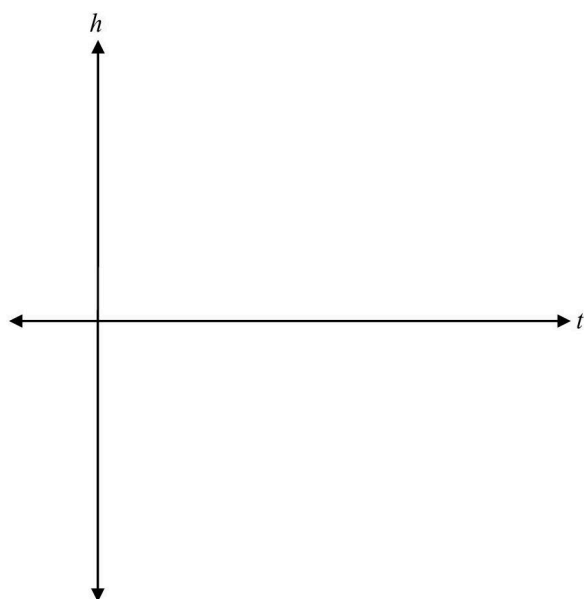
Left-Hand Side	Right-Hand Side	
$2 \cos^2 \left(\frac{\pi}{6} \right)$	$\sin \left(\frac{\pi}{6} \right) + 1$	$\frac{1}{2}$ mark for substitution
$2 \left(\frac{\sqrt{3}}{2} \right)^2$	$\frac{1}{2} + 1$	1 mark for exact values ($\frac{1}{2}$ mark for each)
$2 \left(\frac{3}{4} \right)$		
$\therefore \text{LHS} = \text{RHS}$		2 marks

Question 32

The height of a fish jumping out of the water can be modelled by the function

$h(t) = -t(t-1)(t-4)(t-5)$ where $h(t)$ is the height of the fish above or below the water in cm , and t is the time in seconds, $t \geq 0$.

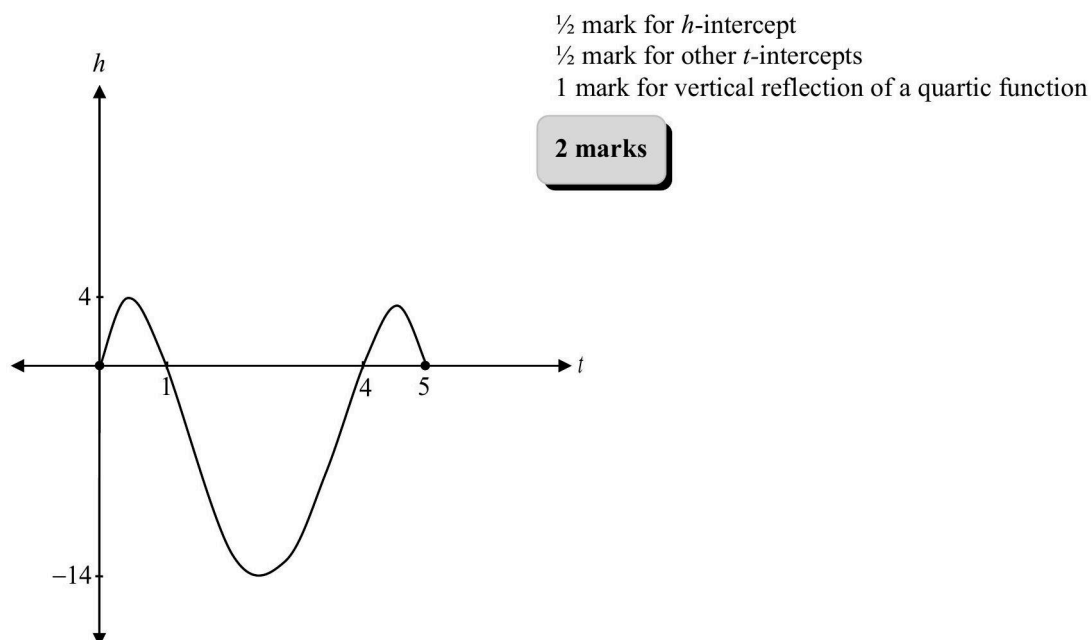
a) Sketch a graph representing the height of the fish with respect to time over the interval $[0, 5]$.



b) State, from the graph in a), the total number of seconds that the fish is above the water.

Solution

a)



b) 2 seconds 1 mark for time consistent with graph in a)

1 mark

Note:

- Scale values on h -axis are not required.

Question 33

Describe the behaviour of the graph of $y = 5^x + 4$ as it approaches $y = 4$.

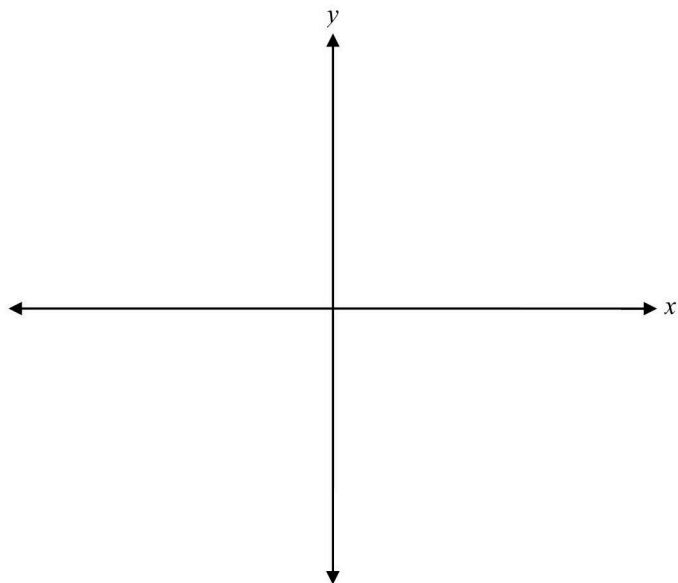
Solution

There is a horizontal asymptote at $y = 4$, so the graph approaches $y = 4$ without ever touching it.

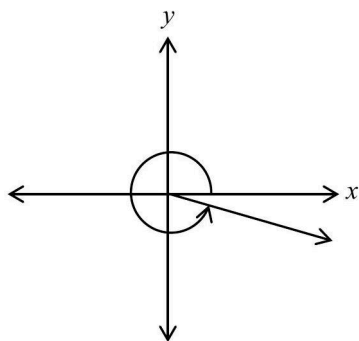
1 mark

Question 34

Sketch the angle of 6 radians in standard position.



Solution



$\frac{1}{2}$ mark for an angle in quadrant IV

$\frac{1}{2}$ mark for an appropriate angle

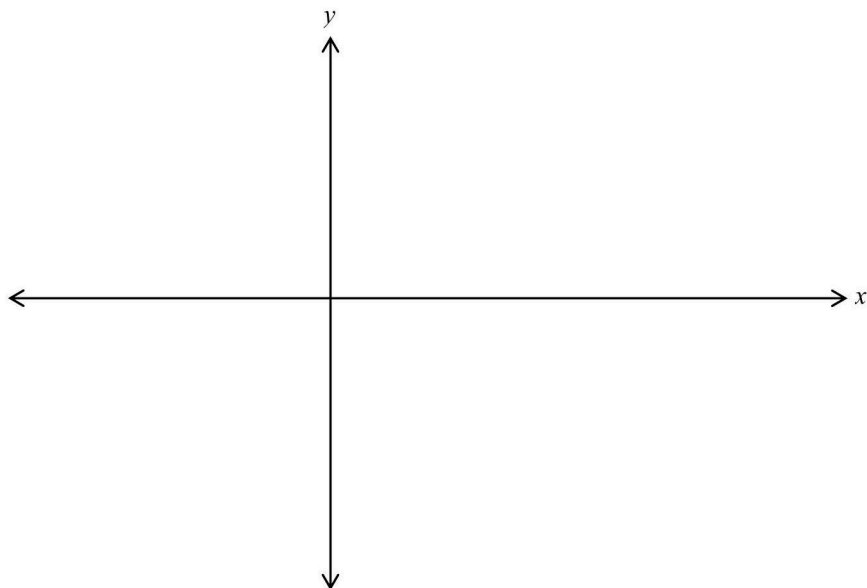
1 mark

Note:

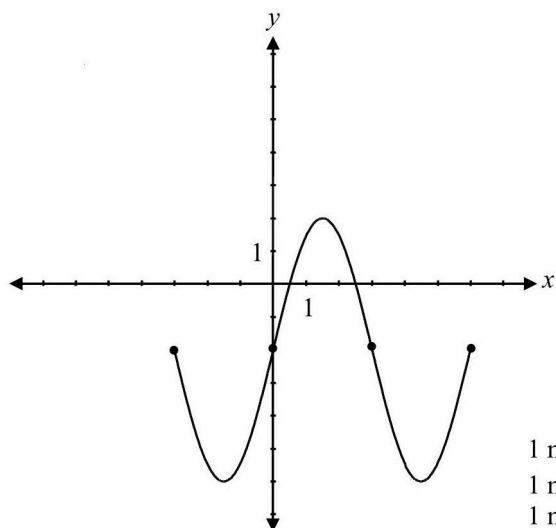
- If the directional arrow is not indicated, deduct an E1 error (final answer not stated).

Question 35

Sketch the graph of the function $y = 4 \sin\left(\frac{\pi}{3}x\right) - 2$ over the domain $[-3, 6]$.



Solution



1 mark for shape of $y = \sin x$
1 mark for amplitude
1 mark for period
1 mark for vertical translation

4 marks

Note:

- Deduct $\frac{1}{2}$ mark for procedural error for not completing the domain of $[-3, 6]$.
- If period mark not awarded, do not deduct for E9 error (scale value on x -axis not indicated).

Question 36

Given $f(x) = 3x - 12$ and $g(x) = x - 4$,

a) determine the equation of $h(x) = \left(\frac{f}{g}\right)(x)$.

$$h(x) =$$

b) describe what the non-permissible value represents on the graph of $h(x)$.

Solution

a) $h(x) = \frac{3x-12}{x-4}$

$$h(x) = \frac{3(x-4)}{(x-4)}$$

$$h(x) = 3, x \neq 4$$

b) The non-permissible value is a point of discontinuity (hole).

1 mark for division of $\left(\frac{f}{g}\right)(x)$

1 mark

1 mark

Question 37

Determine an equation of a radical function, $f(x)$, with a domain of $x \geq 5$ and a range of $y \geq -2$.

$$f(x) =$$

Solution

$$f(x) = \sqrt{x - 5} - 2$$

1 mark for horizontal translation

1 mark for vertical translation

2 marks

- Other equations are possible.

Question 38

Determine the exact value of $\cos\left(\frac{17\pi}{12}\right)$.

Solution

$$\begin{aligned}\cos\left(\frac{17\pi}{12}\right) &= \cos\left(\frac{3\pi}{4} + \frac{2\pi}{3}\right) \\ &= \cos\frac{3\pi}{4} \cos\frac{2\pi}{3} - \sin\frac{3\pi}{4} \sin\frac{2\pi}{3} \\ &= \left(-\frac{\sqrt{2}}{2}\right)\left(-\frac{1}{2}\right) - \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) \\ &= \frac{\sqrt{2}}{4} - \frac{\sqrt{6}}{4} \\ &= \frac{\sqrt{2} - \sqrt{6}}{4}\end{aligned}$$

1 mark for substitution into correct identity

2 marks for exact values ($\frac{1}{2}$ mark for each)

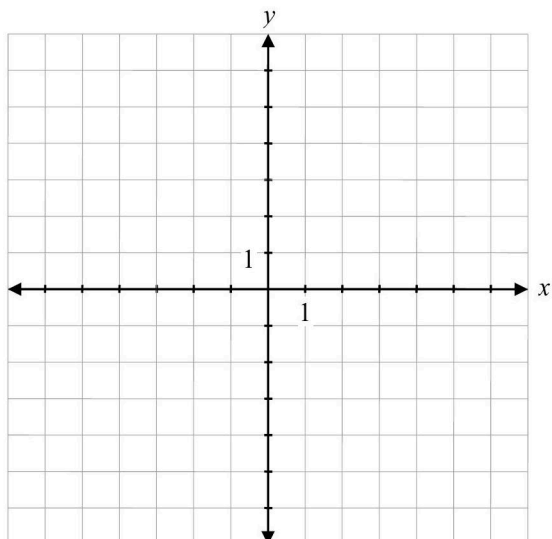
3 marks

Note:

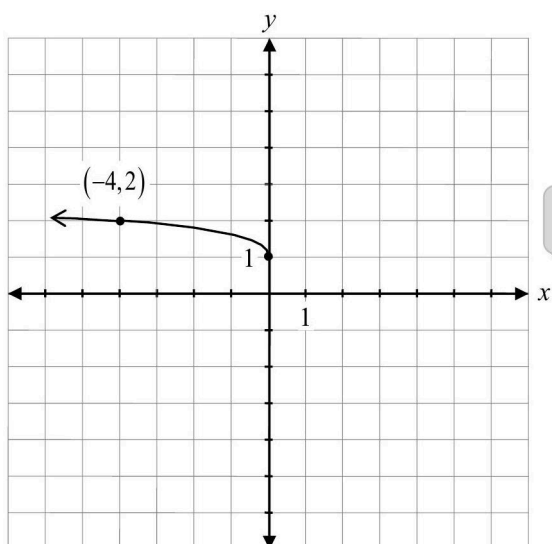
- Other combinations are possible.
- Deduct a maximum $\frac{1}{2}$ mark for arithmetic errors in simplification.

Question 39

Sketch the graph of $y = \frac{1}{2}\sqrt{-x} + 1$.



Solution



1 mark for shape of a radical function

1 mark for vertical compression

1 mark for horizontal reflection

1 mark for vertical translation

4 marks

Question 40

Given $f(x) = \frac{3x}{4} + 9$, determine the equation of $f^{-1}(x)$.

Solution

Let $y = f(x)$

$$y = \frac{3x}{4} + 9$$

$$x = \frac{3y}{4} + 9 \quad 1 \text{ mark for switching } x \text{ and } y \text{ values}$$

$$4x = 3y + 36$$

$$3y = 4x - 36$$

$$y = \frac{4}{3}x - 12 \quad 1 \text{ mark for solving for } y$$

$$f^{-1}(x) = \frac{4x}{3} - 12 \quad 1 \text{ mark for writing equation of } f^{-1}(x)$$

2 marks

Question 41

Describe the end behaviour of the polynomial function $p(x) = -(x - 2)(x + 3)^2$.

Solution

The graph rises as x approaches negative infinity and falls as x approaches positive infinity.

1 mark

Question 42

Given $\csc \theta = -4$ and θ is in quadrant IV,

a) determine the exact value of $\cos \theta$.

b) determine the exact value of $\cot \theta$.

Solution

a) $\sin \theta = -\frac{1}{4}$ 1 mark for reciprocal

$$1^2 - \left(-\frac{1}{4}\right)^2 = \cos^2 \theta$$

$$\frac{15}{16} = \cos^2 \theta$$

$$\pm \frac{\sqrt{15}}{4} = \cos \theta$$

$$\cos \theta = \frac{\sqrt{15}}{4} \quad \left(\begin{array}{l} 1 \text{ mark for } \cos \theta \\ \frac{1}{2} \text{ mark for quadrant; } 1/2 \text{ mark for value} \end{array} \right)$$

2 marks

b)

$$\begin{aligned} \cot \theta &= \frac{\frac{\sqrt{15}}{4}}{-\frac{1}{4}} \\ &= -\sqrt{15} \end{aligned}$$

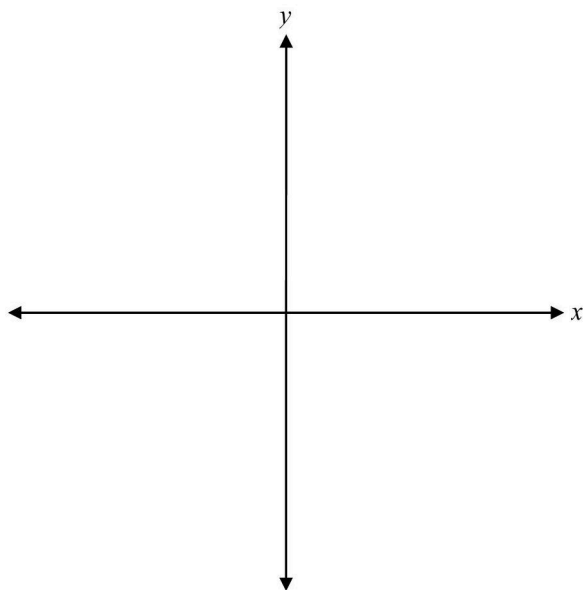
1 mark for $\cot \theta$ consistent with answer in a)

($\frac{1}{2}$ mark for quadrant; $\frac{1}{2}$ mark for value)

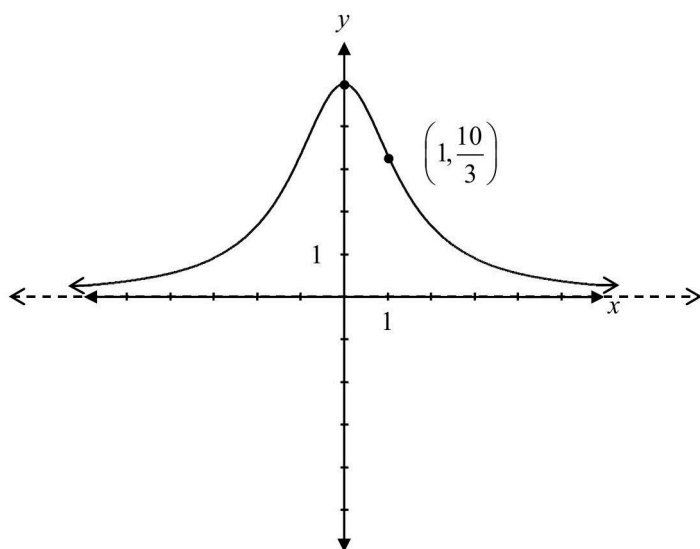
1 mark

Question 43

Sketch the graph of the function $f(x) = \frac{10}{x^2+2}$.



Solution



1 mark for asymptotic behaviour approaching $y = 0$

1 mark for shape ($\frac{1}{2}$ mark for graph left of y -axis; $\frac{1}{2}$ mark for graph right of y -axis)

2 marks

Question 44

Explain why only one of the following equations could be solved algebraically without using logarithms.

$$3^{5x} = 6^{2x-1} \quad \text{or} \quad 16^{2x+3} = \left(\frac{1}{2}\right)^{4x-5}$$

Solution

The equation $16^{2x+3} = \left(\frac{1}{2}\right)^{4x-5}$ can be solved without the use of logarithms because 16 and $\frac{1}{2}$ can be changed to a common base of 2.

Question 45

Given a graph of $y = f(x)$, describe how to sketch the graph of $y = |f(x)|$.

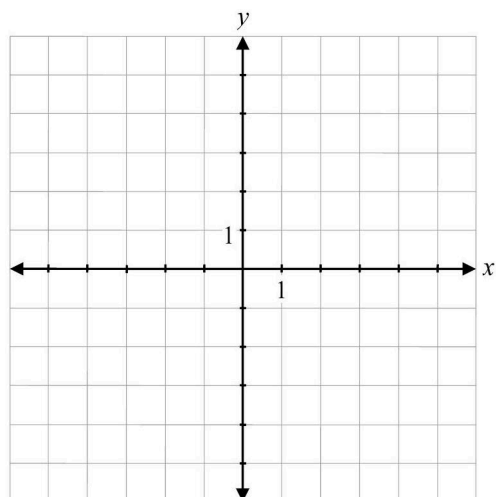
Solution

Reflect all the points with negative y -values over the x -axis.

1 mark

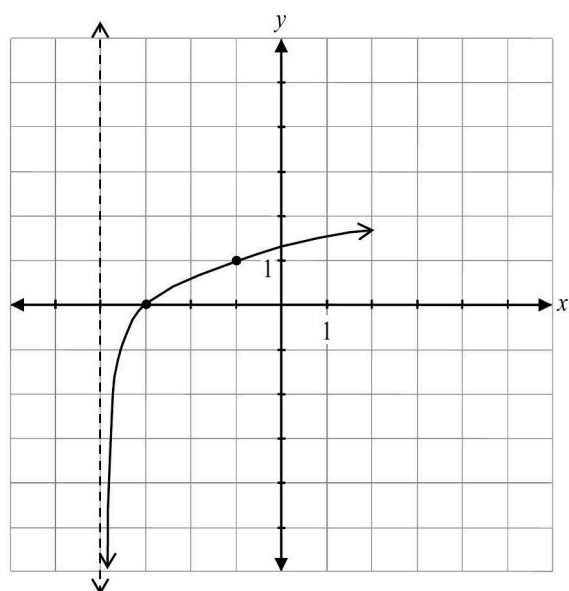
Question 46

Sketch the graph of $y = \log_3(x + 4)$.



Solution

Solution



1 mark for increasing logarithmic function

1 mark for asymptotic behaviour

approaching $x = -4$

2 marks

Question 47

Solve, algebraically.

$$\log x + \log 4 - \log(x - 2) = \log 5$$

Solution

Method 1

$$\log\left(\frac{4x}{x-2}\right) = \log 5$$

$$\frac{4x}{x-2} = 5$$

$$4x = 5(x-2)$$

$$4x = 5x - 10$$

$$10 = x$$

1 mark for product law

1 mark for quotient law

$\frac{1}{2}$ mark for equating arguments

$\frac{1}{2}$ mark for solving for x

3 marks

Method 2

$$\log x + \log 4 - \log(x - 2) - \log 5 = 0$$

$$\log\left(\frac{4x}{5(x-2)}\right) = 0$$

$$10^0 = \frac{4x}{5(x-2)}$$

$$5(x-2) = 4x$$

$$5x - 10 = 4x$$

$$x = 10$$

1 mark for product law

1 mark for quotient law

$\frac{1}{2}$ mark for converting to exponential form

$\frac{1}{2}$ mark for solving for x

3 marks

Question 48

Given $\sin \theta = \frac{1}{2}$, determine all possible values of θ over the interval $[-2\pi, 2\pi]$.

Solution

$$\theta = -\frac{11\pi}{6}, -\frac{7\pi}{6}, \frac{\pi}{6}, \frac{5\pi}{6}$$

1 mark for positive values of θ ($\frac{1}{2}$ mark for each)

1 mark for negative values of θ ($\frac{1}{2}$ mark for each)

2 marks